



HACETTEPE UNIVERSITY

DEPARTMENT OF ELECTRICAL ELECTRONICS ENGINEERING

ELE 489: FUNDAMENTALS OF MACHINE LEARNING

HOMEWORK2

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Here is my github link = <https://github.com/AksuGaye/ELE489-Fundamentals-of-Machine-Learning-HW2>

Q1.

ID	Weather	Wind	Play Outside?
1	Sunny	Strong	No
2	Sunny	Strong	No
3	Overcast	Weak	Yes
4	Rainy	Weak	Yes
5	Rainy	Strong	No
6	Overcast	Strong	Yes

i. Calculate the Gini index for the entire dataset.

Class	Number
Yes	3
No	3
	Total=6

$$Gini\ Index = 1 - \sum_{i=0}^{c-1} p_i(t)^2$$

$$Gini\ Index(entire\ dataset) = 1 - \left(\frac{3}{6}\right)^2 - \left(\frac{3}{6}\right)^2 = 0.5$$

ii. Compute the weighted Gini index for splitting on weather.

Weather	Number of (YES,NO)
Sunny	(0,2)
Overcast	(2,0)
Rainy	(1,1)

$$Gini\ Index(for\ sunny) = 1 - \left(\frac{1}{2}\right)^2 - \left(\frac{1}{2}\right)^2 = 0$$

$$Gini\ Index(for\ overcast) = 1 - \left(\frac{1}{2}\right)^2 - \left(\frac{1}{2}\right)^2 = 0$$

$$Gini\ Index(for\ rainy) = 1 - \left(\frac{0}{2}\right)^2 - \left(\frac{2}{2}\right)^2 = 0.5$$

$$GINI_{split} = \sum_{i=0}^k \frac{n_i}{n} GINI(i)$$

$$GINI_{split} = \frac{2}{6} \cdot 0 + \frac{2}{6} \cdot 0 + \frac{2}{6} \cdot 0.5 = 0.167$$

iii. Compute the weighted Gini index for splitting on wind.

Wind	Number of (YES,NO)
Weak	(2,1)
Strong	(1,2)

$$Gini\ Index(for\ weak) = 1 - \left(\frac{2}{3}\right)^2 - \left(\frac{1}{3}\right)^2 = 0.444$$

$$Gini\ Index(for\ strong) = 1 - \left(\frac{1}{3}\right)^2 - \left(\frac{2}{3}\right)^2 = 0.444$$

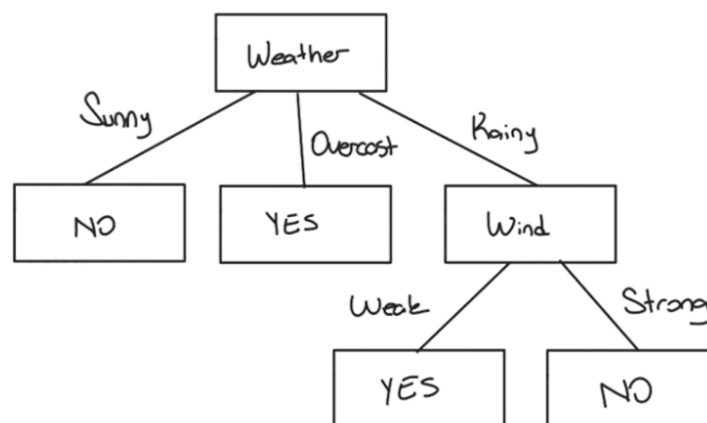
$$GINI_{split} = \sum_{i=0}^k \frac{n_i}{n} GINI(i)$$

$$GINI_{split} = \frac{3}{6} \cdot 0.444 + \frac{3}{6} \cdot 0 = 0.444$$

iv. Choose the feature with the lowest weighted Gini index as the root node.

Attribute	Weighted Gini Index
Weather	0.167
Wind	0.444

This is the weighted impurity of child nodes. Therefore we should choose the attribute that minimizes the weighted average Gini index of the children. **Weather** should be chosen as the root node.



Q2.

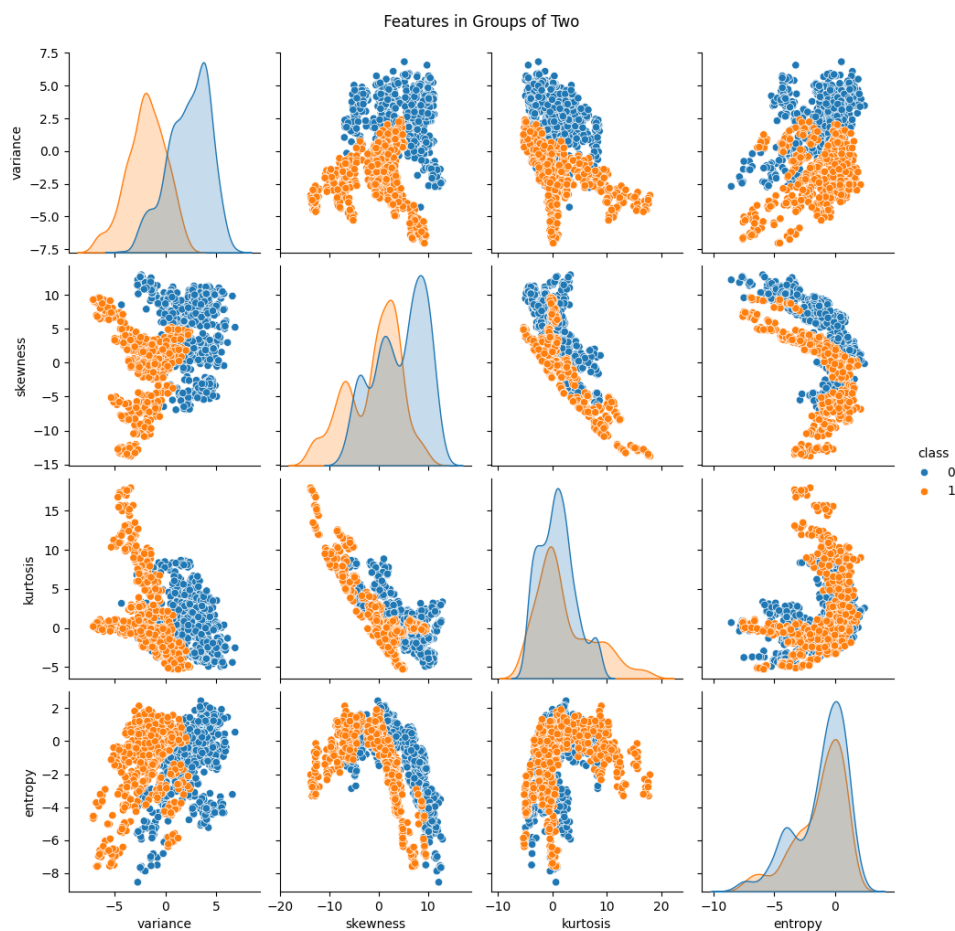
1. Study how to compute an image's variance, skewness, kurtosis, and entropy. What do they measure? Report.

Variance measures how much the pixel values change in the image. The greater the variance, the higher the contrast of the image. Skewness shows if the pixel values are more on the dark side or the bright side. Kurtosis tells us about the shape of the distribution of

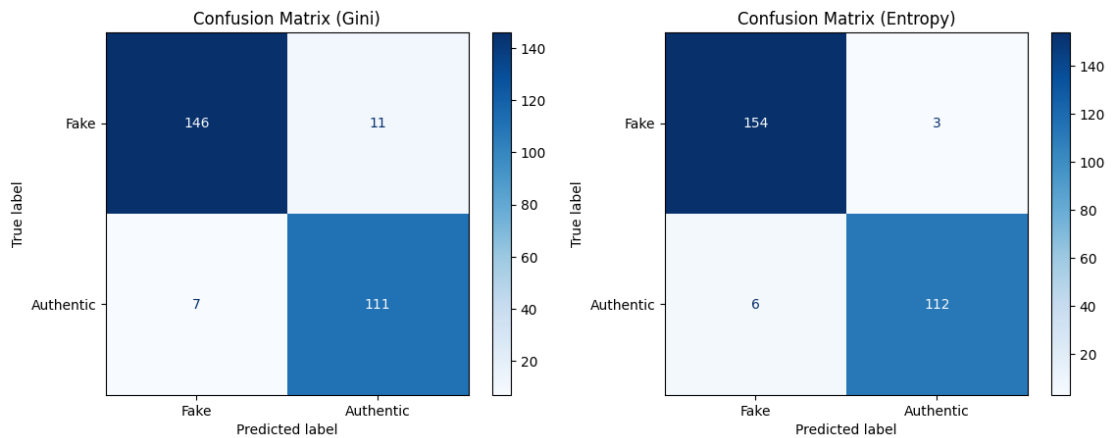
pixels, more specifically about how sharp or flat it is. Entropy measures the amount of information or randomness in the image. A high entropy value means the image has more details or complexity. These values help in understanding the texture and structure of an image for analysis or classification tasks.

2. Do you think the decision tree is a good algorithm to run for these features? Comment.

Yes, I think a decision tree is an appropriate algorithm for this dataset. All four features (variance, skewness, kurtosis, and entropy) are continuous and have no missing values, which is good for training. The class label is binary that is 0 for fake and 1 for authentic. When we look at the pairwise plots, the two classes seem to be separated, and the class boundaries look straightforward. Decision trees use conditions like “feature > value” to split data, which fits this dataset well. They are also easy to understand and do not need scaling. So, a decision tree is a suitable model here.



3. According to the confusion matrices, both models classified most of the samples correctly, but there were fewer mistakes made by the entropy-based model. The Gini model misclassified 18 samples, while the entropy model only misclassified 9. When we look at the classification reports, we can see that the entropy model achieved slightly higher accuracy, precision, recall, and F1-score values. Specifically, the Gini model had an accuracy of 93.45%, while the entropy model reached 96.73%. These results show that both models are good but the entropy-based decision tree provides better classification performance on this data set.



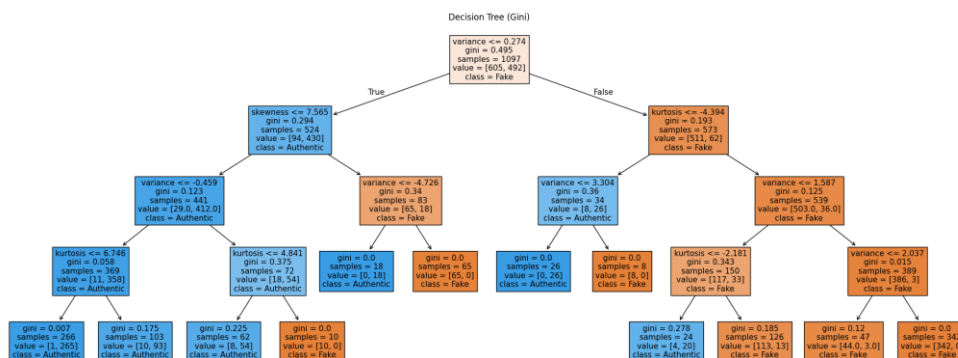
... Classification Report (Gini):

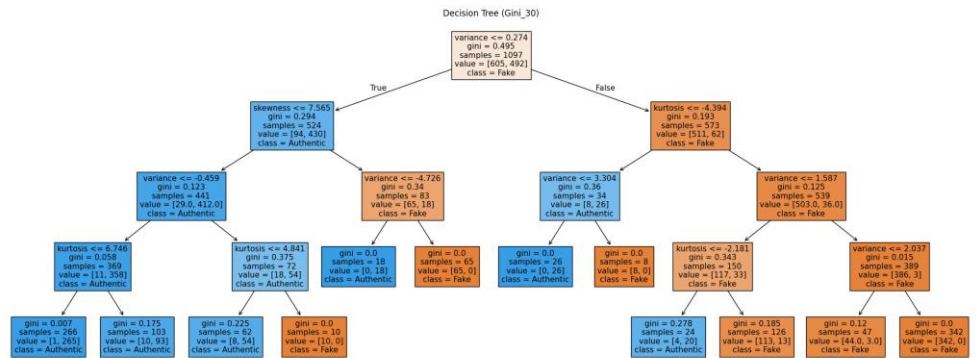
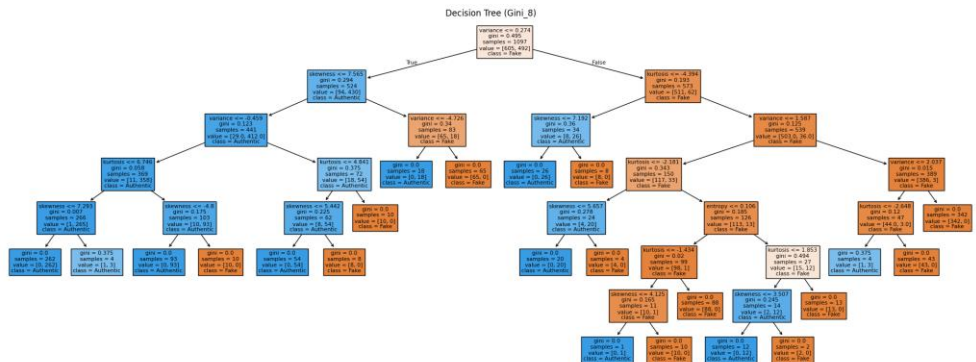
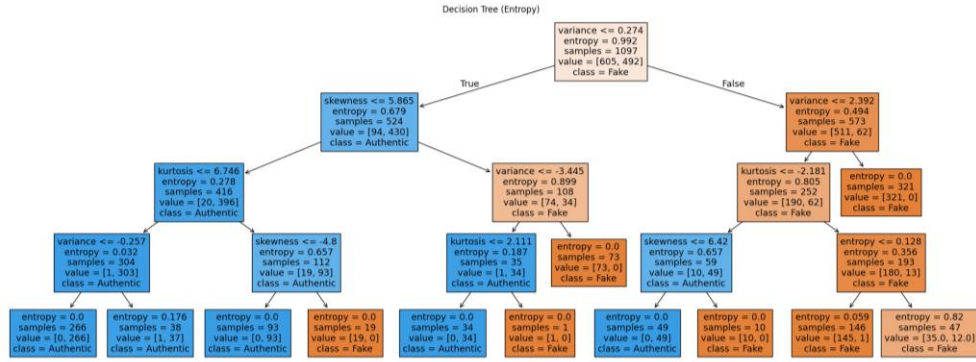
	precision	recall	f1-score	support
0	0.9542	0.9299	0.9419	157
1	0.9098	0.9407	0.9250	118
accuracy			0.9345	275
macro avg	0.9320	0.9353	0.9335	275
weighted avg	0.9352	0.9345	0.9347	275

Classification Report (Entropy):

	precision	recall	f1-score	support
0	0.9625	0.9809	0.9716	157
1	0.9739	0.9492	0.9614	118
accuracy			0.9673	275
macro avg	0.9682	0.9650	0.9665	275
weighted avg	0.9674	0.9673	0.9672	275

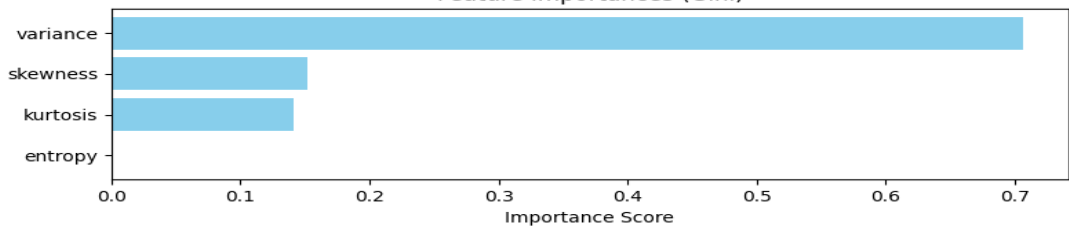
- There are decision trees for different parameters. The Gini and Entropy trees whose both with max_depth=4 look similar, but the Entropy tree is slightly more balanced. If we change max_depth to 8, the tree is considerably larger and harder to read. In contrast, setting min_samples_split=30 limits the tree growth, creating a simpler structure.



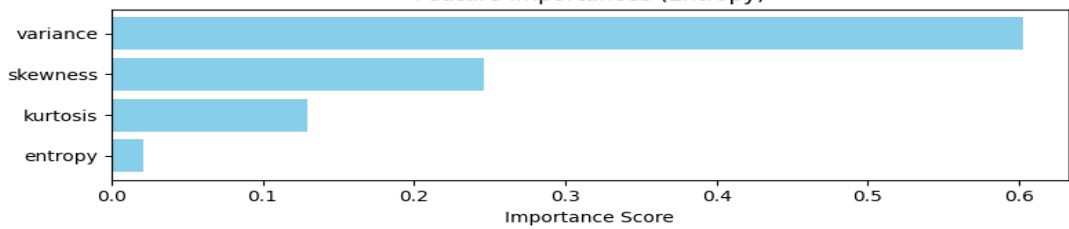


5.

Feature Importances (Gini)



Feature Importances (Entropy)



There are the graphs that show the importance of features for various models. In both the models, variance is most important. However, the Entropy model gives more balanced importance to skewness and kurtosis, while the Gini model relies more heavily on variance.

6. Comment on what you learned from this question. Do you still think decision tree is a good/bad model for this dataset? Why?

From this homework, I learned how to train, evaluate, and visualize a decision tree classifier. I also learned how the tree uses different features and how tree depth affects the complexity of the model. From the confusion matrices, accuracy, and the feature importances, I saw that the decision tree gives good performance on this dataset. The features are numerical and well separated, and the decision boundaries are straightforward, which fits the decision tree model. And I still think a decision tree is a good model for this dataset. It is easy to understand, does not need scaling, and gives high accuracy.